

Technical Architecture and Design Framework for an Adaptive Multi-Modal AI E-Waste Pneumatic Sorting Digital Twin

1. Regulatory Directives, Material Taxonomy, and Target Economics

Automated material recovery systems for Waste Electrical and Electronic Equipment (WEEE) are governed by strict legal mandates and volatile commodity markets. In the European Union, Directive 2012/19/EU (WEEE Recast) establishes a mandatory collection target of 65% of electrical and electronic equipment (EEE) placed on the market, or 85% of total WEEE generated¹. Annex VII of Directive 2012/19/EU requires selective treatment of hazardous and recyclable components prior to mechanical shredding or bulk processing². De-pollution mandates require the complete extraction of printed circuit boards (PCBs) with a surface area exceeding 10 cm^2 ⁴, lithium-ion and lead-acid batteries⁴, cathode-ray tubes (CRTs)³, mercury-containing backlights and switches², brominated flame retardant (BFR) plastics⁶, chlorofluorocarbons/hydrofluorocarbons (CFCs/HFCs)³, external electric cables³, and electrolyte capacitors exceeding 25 mm in height or diameter³.

Simultaneously, the European Union Critical Raw Materials Act enforces processing and recycling targets for strategic raw materials—including neodymium, dysprosium, cobalt, copper, and platinum group metals—to reduce external supply dependencies⁷. In India, the Central Pollution Control Board (CPCB) enforced the E-Waste (Management) Rules 2022, expanding regulated categories from 22 to over 130 product categories⁸. Under these rules, Extended Producer Responsibility (EPR) mandates escalating collection and recycling targets scaling up to 70%–80%⁸. Non-compliance triggers legal action under Environmental Compensation (EC) Regimes: Regime 1 imposes financial compensation based on EPR target shortfalls, while Regime 2 enforces escalating penalties ranging from ₹15,000 to ₹80,000 per violation for operational defaults and improper hazardous waste handling⁸.

To satisfy these regulatory directives while maintaining economic viability on high-speed industrial conveyors ($1.0\text{ to }4.0\text{ m/s}$), an optical pneumatic sorting system must accurately categorize shredded and intact WEEE fragments. The system targets six core material categories, detailed in the table below:

Target	Recovery	Contaminatio	Key Physical &	Physical
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Category	Value (€/kg)	n Risk & Regulatory Severity	Optical Markers	Density & Surface Properties
High-Grade PCBs	€15.00 – €45.00	High (Annex VII mandate; contains Pb, Cd, and halogenated flame retardants) ⁴	Green/Blue/Red solder mask (RGB), strong NIR specular reflections from Au/Cu trace geometry, 1200 nm FR-4 epoxy absorption ⁹	Density: 1.8–2.6 g/cm³ ; Rigid planar shape; High coefficient of friction
IC Molds & Chips	€50.00 – €180.00	Medium (Rich in Au, Ag, Ta, and Rare Earth Elements; high economic priority) ⁷	Matte black surface (RGB absorption), distinct diffusely scattering silica-epoxy resin bands at 1100–1650 nm	Density: 1.6–2.2 g/cm³ ; Small rectangular geometry; Moderate mass
Battery Pack Units	€5.00 – €12.00	Severe (Explosive hazard in shredders; strict Annex VII isolation mandate) ⁴	Cylindrical/pouch metallic casing, thermal heating hysteresis in LWIR, distinct branding labels	Density: 2.2–3.5 g/cm³ ; High mass-to-surface-area ratio; Low belt drag
BFR Polymers	-€0.20 – €0.05	Critical (RoHS/Annex VII hazardous waste; cross-contaminates clean	Dark or grey opaque polymer shell; Sharp vibrational overtone	Density: 1.05–1.40 g/cm³ ; Rigid/semi-rigid shell; Subject to

		plastic streams) ⁶	absorption peaks at 1600–1750 nm	aerodynamic lifting
Metallic Heat Sinks	€1.80 – €6.50	Low (Al/Cu structural scrap; highly valuable secondary non-ferrous stream)	High surface metallic luster (RGB), uniform broadband NIR reflectance (>), high thermal conductivity ⁹	Density: 2.70–8.96 g/cm ³ ; High mass; Complex finned aerodynamic drag profile
Copper Wire Harnesses	€4.50 – €7.20	Low (Annex VII requires removal of external cables prior to processing) ³	Flexible linear geometry, textured PVC/PE insulation jackets (RGB/NIR), exposed copper core cross-sections	Density: 1.3–3.0 g/cm ³ ; Non-rigid/deformable; High probability of inter-item tangling

2. Multi-Modal Optical Physics and Pre-trained Model / Dataset Strategy

Multispectral Perception Physics

Standard 3-channel RGB computer vision is insufficient for industrial e-waste sorting. RGB sensors fail when materials exhibit identical visual coloration (such as black BFR-containing Acrylonitrile Butadiene Styrene [ABS] vs. non-BFR Polycarbonate [PC]), or when surfaces are contaminated with industrial dust, soot, or oxidation. A multi-modal optical sensing architecture resolves these ambiguities by capturing distinct wavebands across the electromagnetic spectrum:

- **Visible RGB (400–700 nm)**: Captures high-spatial-resolution contextual information, surface printing, component boundaries, and primary color profiles⁹. However, RGB perception is vulnerable to specular glare and surface contamination.
- **Near-Infrared (NIR: 750–1000 nm)**: Glass-reinforced epoxy laminates (FR-4 substrates) absorb specific NIR wavelengths through hydroxyl (O-H) and

carbon-hydrogen (C-H) vibrational harmonics. Conversely, metallic traces (gold plating, exposed copper, tin-lead solders) act as near-perfect specular reflectors across the NIR band⁹. Integrating a 4th-channel NIR tensor allows the system to segment fine conductive traces from background insulating boards under variable illumination.

- **Short-Wave Infrared (SWIR: 1000–2500 nm):** Polymer identification relies on fundamental molecular bond resonances. Brominated Flame Retardants (BFRs) alter the spectral reflectance profile of polymers, producing sharp attenuation bands between 1600 nm and 1750 nm due to C-Br and altered C-H stretch overtones, distinguishing toxic polymers from clean recyclable fractions⁶.
- **Long-Wave Infrared (LWIR: 8000–15000 nm):** Thermal imaging measures differential radiative cooling and thermal emissivity. Heated metallic components (such as aluminum heat sinks) exhibit low emissivity ($\epsilon \approx 0.05\text{--}0.2$) and rapid thermal dissipation, whereas polymer casings exhibit high emissivity ($\epsilon \approx 0.90\text{--}0.95$) and retain heat longer, providing clear separation during post-shredding thermal excitation¹³.

Pre-trained Vision Weight Audit and Raw Dataset Fallback Topology

Building an edge-deployable multi-modal neural network requires balancing parameter size against inference latency. The system leverages pre-trained vision foundations—specifically Segment Anything Model 2 (SAM2), DINOv2 ViT backbones¹³, and lightweight YOLOv8/YOLOv11-seg feature extractors pre-trained on COCO/ImageNet¹⁴—adapting them to 4-channel input domains via transfer learning.

When pre-trained weights for specialized 4-channel multi-modal inputs are absent, raw benchmark datasets serve as fallbacks for supervised fine-tuning:

- **PCB-Vision Benchmark:** A high-resolution dataset containing 53 multi-scene co-registered RGB and hyperspectral imaging (HSI, 400–1000 nm) acquisitions captured via a 4096-pixel RGB CMOS line-rate camera paired with a Specim FX10 push-broom sensor⁹. It provides pixel masks for component-level PCB sorting⁹.
- **SpectralWaste Benchmark:** Comprises 852 labeled scenes and 6,803 unlabeled scenes of co-registered RGB and HSI (1000–1700 nm , Specim FX17) data⁹. Across six target classes (cardboard, film, basket, filaments, video tape, trash bag)¹⁷, state-of-the-art models such as Bidirectional Cross-Attention Fusion (BCAF) achieve 75.4%–76.4% mIoU at frame rates ranging from 31 to 55 FPS¹¹, while COSNet achieves 69.96% mIoU on RGB-only subsets¹⁷.
- **DeepPCB & PKU-Market-PCB Datasets:** DeepPCB contains 1,500 aligned image pairs covering six defect classes (open, short, mousebite, spur, copper, pin-hole)¹⁹. PKU-Market-PCB offers 3,505 augmented images annotated for micro-defect instance

segmentation¹⁹. These datasets allow pre-training of spatial feature extractors on fine PCB geometry before deployment on multi-modal streams¹⁹.

- **Electrolyzers-HSI Dataset:** Contains 55 multi-scene co-registered RGB and HSI data cubes ($380\text{--}2500\text{ nm}$) covering mesh, steel anodes/cathodes, and high-temperature electrolyzer components⁹. Models evaluated on this dataset (such as Multimodal Fusion Transformers [MFT]) demonstrate the performance gains of combining spatial RGB structures with continuous spectral channels⁹.
- **ThermalRGBTrash Benchmark:** Provides 22 co-registered RGB-LWIR video streams containing 901 labeled tracklets and 5,429 instance masks¹³. It serves as the baseline for evaluating cross-spectral transfer learning under thermal noise¹³.

Industrial Data Augmentation via Albumentations

To handle the optical noise of a Material Recovery Facility (MRF), the multi-modal input pipeline employs an Albumentations transformation sequence during fine-tuning. Raw image tensors pass sequentially through directional motion blur modeling, specular glare injection, industrial dust masking, and CutOut occlusions to generate robust feature representations:

Directional motion blur simulates high conveyor belt speeds ($1.0\text{ to }4.0\text{ m/s}$) by applying a linear convolution kernel K_{blur} aligned along the belt movement axis $\theta \in [-2^\circ, +2^\circ]$, with kernel size $k \in [15, 31]\text{ pixels}$:

$$K_{\text{blur}}(x, y) = \begin{cases} \frac{1}{k} & \text{if } y = x \tan \theta \text{ and } |x| \leq \frac{k}{2} \\ 0 & \text{otherwise} \end{cases}$$

Specular glare and highlight injection simulates directional lighting over reflective metals by superimposing high-intensity elliptical Gaussian spots

$I_{\text{glare}} = A \exp \left(- \left(\frac{(x-x_0)^2}{2\sigma_x^2} + \frac{(y-y_0)^2}{2\sigma_y^2} \right) \right)$ onto the RGB and NIR channels, saturating target pixel intensities to 255. Industrial dust and dirt masking applies multi-scale Perlin noise maps to modulate channel opacity, recreating particulate deposition on optical lenses and conveyor belts. Finally, random occlusion and tangling via CutOut or Random Erasing removes rectangular and irregular polygonal regions (up to 30% of total target area), forcing the network to infer instance boundaries from partially visible fragments.

3. Model Architecture, Weight Adaptation, and Edge Optimization

4-Channel Tensor Adaptation Architecture

Integrating a 4th channel (NIR or Infrared) into a standard 3-channel RGB vision backbone (such as YOLOv8-seg or YOLOv11-seg) without destroying pre-trained COCO/ImageNet

spatial features requires structural tensor adaptation¹⁴. Directly resetting the first convolutional layer (Conv1) to random uniform initialization destroys low-level feature detectors (edge detection, color gradients, texture primitives), increasing fine-tuning convergence time. Instead, the input tensor is split into RGB channels and the auxiliary 4th channel. The first layer's

weight tensor $\mathbf{W}_{\text{conv1}} \in \mathbb{R}^{C_{\text{out}} \times 3 \times K_h \times K_w}$ is expanded to

$\mathbf{W}_{\text{conv1}}^{(4)} \in \mathbb{R}^{C_{\text{out}} \times 4 \times K_h \times K_w}$ via Weight Inflation:

$$\mathbf{W}_{\text{conv1}}^{(4)}[:, 0 : 3, :, :] = \mathbf{W}_{\text{RGB}}$$

$$\mathbf{W}_{\text{conv1}}^{(4)}[:, 3, :, :] = \frac{1}{3} \sum_{c=0}^2 \mathbf{W}_{\text{RGB}}[:, c, :, :] \quad \text{or} \quad \mathbf{0}$$

Initializing the 4th channel with the mean activation weight of the RGB channels preserves initial feature magnitude, preventing gradient instability in the early steps of fine-tuning. For multi-stream architectures (such as BCA^{F11}), spatial RGB representations and lower-resolution NIR/HSI slices pass through separate encoders. Cross-attention layers align features across scales without collapsing spectral detail:

$$\text{Attention}(Q_{\text{RGB}}, K_{\text{NIR}}, V_{\text{NIR}}) = \text{softmax} \left(\frac{Q_{\text{RGB}} K_{\text{NIR}}^T}{\sqrt{d_k}} \right) V_{\text{NIR}}$$

This enables localized cross-modal interaction between high-resolution spatial edges (RGB) and material spectral signatures (NIR)¹¹.

Fine-Tuning Strategy and Composite Loss Formulation

Fine-tuning employs a two-stage training schedule:

1. **Stage 1 (Feature Extraction Freeze):** The backbone layers are frozen for the first 10 epochs. Only the newly inflated Conv1 4th-channel weights and segmentation head projections are updated using AdamW ($\text{lr} = 1 \times 10^{-3}$).
2. **Stage 2 (End-to-End Unfreezing):** All network layers are unfrozen and trained for 100 epochs using Cosine Annealing learning rate scheduling:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos \left(\frac{t}{T_{\max}} \pi \right) \right)$$

where $\eta_{\max} = 1 \times 10^{-4}$, $\eta_{\min} = 1 \times 10^{-6}$, and $T_{\max} = 100$.

The composite loss function balances object detection, bounding box localization, class assignment, and pixel-level instance segmentation:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{box}} \mathcal{L}_{\text{CIoU}} + \lambda_{\text{dfl}} \mathcal{L}_{\text{DFL}} + \lambda_{\text{cls}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{mask}} \mathcal{L}_{\text{Dice}}$$

$$\mathcal{L}_{\text{CIoU}} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \alpha v$$

where $\rho(\cdot)$ is the Euclidean distance between box centerpoints, c is the diagonal length of the smallest enclosing box, $v = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2$, and $\alpha = \frac{v}{(1-\text{IoU})+v}$.

Distribution Focal Loss ($\mathcal{L}_{\text{DFL}}$) optimizes bounding box boundaries under motion blur by modeling box edges as continuous probability distributions. Mask segmentation loss uses a combination of Binary Cross-Entropy ($\mathcal{L}_{\text{BCE}}$) and Dice Loss ($\mathcal{L}_{\text{Dice}}$):

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i p_i y_i + \epsilon}{\sum_i p_i + \sum_i y_i + \epsilon}$$

Edge Acceleration and Quantization Pipeline

To meet high-speed conveyor demands ($\geq 50 \text{ FPS}$, or $\leq 20 \text{ ms}$ per frame execution budget), PyTorch checkpoints are converted to Open Neural Network Exchange (ONNX)

format with dynamic batch sizes and fixed spatial dimensions (640×640 or

1024×1024). The ONNX models are then compiled into optimized TensorRT engines.

Float32 weights and activations are quantized to 8-bit integers using TensorRT's `Int8EntropyCalibrator2`, evaluated over a calibration dataset containing 1,000 multi-modal MRF images. This process minimizes Kullback-Leibler (KL) divergence between FP32 and INT8 activation distributions:

$$D_{\text{KL}}(P \parallel Q) = \sum_{i=1}^N P(i) \log \left(\frac{P(i)}{Q(i)} \right)$$

The execution engine uses dual CUDA streams (cudaStream_t) to overlap host-to-device memory transfers (cudaMemcpyAsync) with GPU tensor computation, reducing processing latency to sub-15 millisecond windows.

4. High-Speed Kinematics, Multi-Object Tracking, and Spatial Calibration

Multi-Object Tracking (MOT) Benchmarking

In high-speed conveyor sorting, tracking-by-detection algorithms must maintain stable identity labels for objects moving at 1.0 to 4.0 m/s . Identity switches (IDSW) cause improper nozzle firing, leading to sorting contamination. The table below benchmarks candidate tracking algorithms under industrial belt conditions:

Tracker Framework	Primary Association Mechanism	MOTA (%)	HOTA (%)	IDF1 (%)	Average FPS	Computational Overhead	Robustness to Conveyor Blur & Occlusion
ByteTrack	Two-stage association matching high and low confidence detections via Kalman Filter & IoU ²²	75.7% [cite: 23]	60.9% [cite: 23]	77.3%	65.0+ [cite: 24]	Ultra-Low (No ReID network required) ²²	High (Recovers occluded/blurred items from low-score detections) ²²
DeepSORT	Kalman filter combined with deep	64.2%	43.7% ²⁴	65.8% ²⁵	10.5 ²⁴	High (Requires crop extraction +	Poor (Motion blur degrades

	appearance ReID feature embeddings ²²					ReID inference per target) ²⁴	appearance embeddings, causing ID switches ²²
BoT-SORT	Camera Motion Compensation (CMC), Kalman filter, and refined ReID features ¹⁴	75.1%	51.3% ¹⁴	75.8%	10.6 ²⁴	Extremely High (CMC optical flow calculation adds latency) ²⁴	Moderate (High accuracy, but throughput is too low for fast conveyors) ²²
OC-SORT	Observation-Centric momentum updates to fix Kalman velocity drift ²²	74.8%	59.8%	76.2%	55.0 ²⁴	Low (Pure motion model with virtual trajectory rewinding) ²⁶	Excellent (Handles non-linear motion when items roll or slip) ²²

DeepSORT and BoT-SORT rely heavily on deep Re-Identification (ReID) feature extractors²². On high-speed conveyors, directional motion blur alters object appearance, degrading cosine similarity matrices and causing frequent identity swaps. Additionally, running a secondary ReID inference pass for every bounding box crop drops frame throughput below **15 FPS**²⁴. ByteTrack overcomes these limitations by eliminating the ReID network entirely²². It retains low-confidence detection boxes ($0.1 < \text{score} \leq 0.5$) rather than discarding them²². On an e-waste conveyor, items undergoing partial occlusion or heavy motion blur experience

transient drops in detector confidence. In the first association stage, ByteTrack matches high-confidence detections ($\text{score} > 0.5$) with active tracklets using Intersection-over-Union (IoU) distance²². Unmatched tracklets are then passed to a second association stage, where they are matched against low-confidence detections²². This recovers occluded or blurred target trajectories without adding computational overhead²². For conveyors where fragments roll or shift position unexpectedly, OC-SORT provides an alternative fallback by replacing standard constant-velocity Kalman assumptions with observation-centric momentum updates²².

2D Planar Homography Transformation

To transform pixel coordinates (u, v) from camera frames into real-world metric coordinates (x_w, y_w) on the conveyor surface, the system uses a 2D Planar Homography matrix $\mathbf{H} \in \mathbb{R}^{3 \times 3}$:

$$\begin{bmatrix} x_w \\ y_w \\ 1 \end{bmatrix} \sim \mathbf{H} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

Before applying homography, raw pixel coordinates are corrected for lens distortion using radial (k_1, k_2, k_3) and tangential (p_1, p_2) coefficients:

$$x_{\text{corrected}} = x(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + [2p_1 xy + p_2(r^2 + 2x^2)]$$

$$y_{\text{corrected}} = y(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + [p_1(r^2 + 2y^2) + 2p_2 xy]$$

where $r^2 = x^2 + y^2$.

The transformation matrix $\mathbf{H}$ is solved via the Direct Linear Transform (DLT) algorithm using $N \geq 4$ non-collinear reference points (such as high-precision ArUco calibration targets embedded along the belt margins)²⁸. The linear system $\mathbf{A}\mathbf{h} = \mathbf{0}$ is formulated as:

$$\mathbf{A}_i = \begin{bmatrix} -u_i & -v_i & -1 & 0 & 0 & 0 & u_i x_{w,i} & v_i x_{w,i} & x_{w,i} \\ 0 & 0 & 0 & -u_i & -v_i & -1 & u_i y_{w,i} & v_i y_{w,i} & y_{w,i} \end{bmatrix}$$

Singular Value Decomposition (SVD) decomposes $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, where the optimal homography vector $\mathbf{h}$ corresponds to the right-singular vector associated with the smallest singular value (the last column of $\mathbf{V}$).

5. Pneumatic Actuation Physics and Microsecond Scheduling Dynamics

Deterministic Event-Loop Scheduling

Pneumatic sorting systems eject target materials using a transverse array of high-speed solenoid air nozzles positioned downstream from the vision camera's field of view. At belt speeds of 4.0 m/s , a timing delay error of 1 millisecond results in a 4.0 mm spatial targeting displacement, missing the item or deflecting adjacent clean material. To prevent operating system thread-preemption jitter, the timing controller uses microsecond-resolution monotonic clock calls (`time.monotonic_ns()`). The system maps an item's world centroid $(x_w(t), y_w(t))$ to a specific nozzle index k within a 64-nozzle manifold spanning a 1.0 m conveyor width (nozzle pitch $\Delta y = 15.625 \text{ mm}$):

$$k = \left\lfloor \frac{y_w(t) + \frac{W_{\text{belt}}}{2}}{\Delta y} \right\rfloor$$

The precise firing timestamp t_{fire} is calculated as:

$$t_{\text{fire}} = t_{\text{detect}} + \left(\frac{x_{\text{nozzle}} - x_w(t_{\text{detect}})}{v_{\text{belt}}} \right) - t_{\text{actuation_lag}}$$

where $t_{\text{actuation_lag}} = t_{\text{solenoid_rise}} + t_{\text{air_travel}}$. For typical industrial solenoid valves, $t_{\text{actuation_lag}} \approx 2.5 \text{ ms} \pm 0.1 \text{ ms}$.

Pneumatic Ejection Physics & Secondary Collision Dynamics

The force exerted by a compressed air nozzle operating at supply pressure P_j ($6.0\text{--}8.0 \text{ bar}$) through an orifice area A_j onto an e-waste fragment of mass m is modeled via compressible flow dynamics:

$$v_j = \sqrt{\frac{2\gamma}{\gamma-1}RT_0 \left[1 - \left(\frac{P_0}{P_j} \right)^{\frac{\gamma-1}{\gamma}} \right]}$$

$$F_{\text{jet}} = \dot{m}v_j + (P_j - P_0)A_j = \rho_j A_j v_j^2 + (P_j - P_0)A_j$$

where $\gamma = 1.4$ for air, R is the specific gas constant, and ρ_j is air density at the nozzle exit.

The minimum required valve open duration Δt_{pulse} to impart an upward deflection velocity $v_{z,\text{req}}$ to overcome gravity and clear the splitter gate is:

$$\Delta t_{\text{pulse}} = \frac{m_{\text{target}} \cdot v_{z,\text{req}}}{F_{\text{jet}} - m_{\text{target}} g}$$

High-pressure air jets create secondary aerodynamic shockwaves and boundary-layer turbulence around the target component. These air currents can displace light, adjacent unselected items (such as thin plastic films or small surface-mount capacitors), causing unintended collateral ejection—a phenomenon known as the Popcorn Effect.

To suppress this effect, the digital twin implements two control strategies:

1. **Dynamic Pulse Modulation:** Valve open duration Δt_{pulse} scales dynamically based on instance mask area A_{mask} extracted by the segmentation network. Light items receive shorter, low-impulse blasts (1.5–3.0 ms), while heavy components (such as heat sinks) receive longer blasts (5.0–10.0 ms).
2. **Spatial Guard-Band Zoning:** If an unselected item lies within a critical spatial radius

$r_{\text{crit}} = \sqrt{\frac{F_{\text{jet}} \Delta t_{\text{pulse}}}{\pi \rho_{\text{air}} v_{\text{threshold}}}}$ of a targeted ejection centroid, adjacent nozzles are suppressed or modulated to prevent cross-contamination.

6. Industrial State-of-the-Art Benchmarking and Zero-Cost Digital Twin Architecture

Commercial Industrial Systems vs. Digital Twin Simulation

To benchmark operational performance, the proposed software digital twin is evaluated against established commercial e-waste optical sorting platforms:

System Parameter	TOMRA AUTOSORT	STEINERT UniSort PR	Pellenc ST COMPACT	REDWAVE X-TRACT	Proposed Digital Twin Simulation
Max Belt Speed	3.0–4.5 m/s	3.0–4.0 m/s	2.5–3.5 m/s	2.0–3.0 m/s	1.0–4.0 m/s (Configurable)
Primary Sensors	NIR + RGB Camera	HSI (NIR/SWIR) + Color	NIR + VIS + Dual-Energy XRT	Transmission X-Ray (XRT) + RGB	4-Channel RGB + NIR / SWIR Fusion
Target Throughput	3.0–8.0 ton	2.5–6.0 ton	2.0–5.0 ton	4.0–10.0 ton	Simulated 1.0–5.0 ton
System Latency	< 10 ms	< 12 ms	< 15 ms	< 8 ms	12.5 ms (≥ 50 FPS Target Execution)
Spatial Resolution	1.0–2.0 mm	2.0 mm	1.5 mm	1.0 mm	0.8 mm/pixel (1024 × 1024 at 0.8 m FOV)
Capital Cost (\$)	\$150,000 – \$350,000+	\$180,000 – \$400,000+	\$140,000 – \$300,000+	\$200,000 – \$450,000+	\$0.00 (Zero-Cost Open Simulation Framework)

Digital Twin System Architecture (Producer-Worker-Consumer Pipeline)

The zero-cost digital twin simulates high-speed sorting on consumer-grade hardware (such as a laptop equipped with an NVIDIA RTX GPU) without physical conveyor hardware. It uses a multi-threaded, process-isolated Producer-Worker-Consumer architecture operating over lock-free shared memory ring buffers.

The software pipeline executes through four concurrent processing stages:

1. **Producer Process (Synthetic Frame Stream Generator):** Reads dual-stream video files or raw synthetic benchmark datasets (such as SpectralWaste or PCB-Vision⁹), generating co-registered 4-channel tensors ($B \times 4 \times 640 \times 640$) at 60 FPS. Tensors are written directly to a shared memory block (SharedMemory) to avoid inter-process serialization overhead.
2. **Worker Process 1 (TensorRT Multi-Modal Inference Engine):** Dequeues frame pointers from the frame queue (FrameQueue). It executes sub-10ms inference using an INT8-quantized TensorRT engine, extracting target class predictions, bounding boxes, and polygon instance masks.
3. **Worker Process 2 (Kinematic Engine & Multi-Object Tracker):** Consumes raw detection outputs from the detection queue (DetectionQueue). It runs ByteTrack to update trajectory identities²², applies the planar homography matrix $\mathbf{H}$ to map coordinates to metric space, projects linear conveyor movement $x_w(t) = x_{w,0} + v_{\text{belt}} \cdot t$, and assigns target nozzle indices k alongside microsecond firing timestamps t_{fire} .
4. **Consumer Process (Actuator Simulator & Telemetry Dashboard):** Monitors the actuation queue (ActuationQueue) using a high-resolution discrete event loop (time.monotonic_ns()). When the system clock reaches t_{fire} , the process simulates pneumatic nozzle pulses, models 2D ballistic ejection paths, tracks sorting purity and yield metrics, and renders a 60 FPS visual telemetry interface.

To maintain real-time throughput above **50 FPS** under heavy conveyor loads without frame drops or memory growth, the ring buffer implements an adaptive queue-throttling policy. If FrameQueue occupancy exceeds 80% capacity, worker nodes dynamically bypass segmentation inference for intermediate frames, relying on Kalman kinematic extrapolation to update active tracklets until queue pressure normalizes.

7. Conclusions and System Operational Roadmap

This study establishes a technical blueprint for deploying edge-accelerated multi-modal vision models within an e-waste pneumatic sorting pipeline. Integrating an auxiliary NIR/Infrared channel alongside standard RGB inputs provides the optical separation required to satisfy EU

WEEE Annex VII de-pollution directives and India CPCB EPR mandates².

Key architectural conclusions include:

- **Tensor Adaptation via Weight Inflation:** Expanding pre-trained 3-channel vision backbones using first-layer weight inflation preserves low-level spatial features learned from COCO/ImageNet while enabling fine-tuning on multi-modal inputs.
- **Tracking Without ReID Overhead:** High-speed sorting pipelines require real-time tracking algorithms that avoid deep ReID appearance feature extraction. ByteTrack's two-stage association mechanism handles motion blur and transient occlusions while maintaining processing throughput above **60 FPS**²².
- **Deterministic Actuation Timing:** Pneumatic ejection at belt speeds up to **4.0 m/s** requires microsecond-level clock scheduling (`time.monotonic_ns()`). Dynamic pulse modulation guided by instance segmentation masks mitigates the secondary popcorn effect, protecting clean material streams from collateral deflection.
- **Validation via Digital Twin:** The multi-threaded Producer-Worker-Consumer architecture simulates real-time conveyor sorting on standard laptop hardware. This zero-cost digital twin enables pipeline optimization prior to physical hardware deployment.

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